

MDI3003 - ADVANCED PREDICTIVE ANALYTICS

Lab - 1 Report

House Price Prediction using Linear Regression

An End-to-End Regression Study on the California Housing Dataset

Course: MDI3003 - Advanced Predictive Analytics

Faculty: Dr. Durgesh Kumar, Assistant Professor (Senior), SCOPE, VIT Vellore

Dataset: California Housing (scikit-learn)

Environment: Python 3, scikit-learn, pandas, NumPy, Matplotlib, Seaborn

Github link: https://github.com/Madhumasa84/Adv_predictive/tree/main/lab1

[Madhusudhanan G]

[23MID0444]

Table of Contents

1. Executive Summary
 2. Business Problem and Objective
 3. Dataset Description
 4. Methodology
 5. Exploratory Data Analysis and Feature Engineering
 6. Model Development
 7. Evaluation Results
 8. Interpretation of Results
 9. Limitations and Responsible-Use Risk
 10. Future Improvements
 11. Conclusion
 12. Discussion Questions (Faculty-Assigned)
 13. Viva Questions (Faculty-Assigned)
- Appendix A: Reproducibility Notes
- References

1. Summary

This report documents an end-to-end regression study carried out for MDI3003 (Advanced Predictive Analytics), Lab 01, on the theme of house price prediction using Linear Regression. The study follows the ten-stage workflow prescribed in the laboratory manual: business understanding, data understanding, holdout design, data preparation, exploratory analysis, baseline modelling, validation and tuning, final evaluation, interpretation, and reproducibility.

The California Housing dataset (20,640 observations, 8 numeric predictors, target = median district house value expressed in units of \$100,000) was used as the analytical basis for this core laboratory submission. An 80/20 train-test split with a fixed random seed (42) was used throughout to ensure reproducibility.

Six models were trained and compared under an identical evaluation protocol: a naive mean-prediction baseline, Simple Linear Regression (single feature), Multiple Linear Regression (all eight features), Ridge Regression (L2-regularized, hyperparameter-tuned), Lasso Regression (L1-regularized, hyperparameter-tuned), and a Random Forest Regressor (tuned via grid search with 5-fold cross-validation).

The tuned Random Forest Regressor produced the best generalization performance, achieving a test RMSE of 0.5077 and a test R^2 of 0.8033 - a 55.7% reduction in RMSE relative to the naive baseline. Median household income (MedInc) was found to be the single strongest linear predictor of house value (correlation 0.69), while the full multiple-regression model and the regularized variants offered comparable, more modest linear performance (test RMSE close to 0.745, R^2 close to 0.576). The study concludes that Random Forest is preferred where predictive accuracy is the priority, while Ridge/Lasso remain preferable where coefficient-level interpretability is required for business communication.

Key limitation: this deliverable uses a single dataset (California Housing) rather than the two-dataset extended-study requirement; it is scoped and presented here as the 3-hour core-laboratory submission with a full six-model comparison, consistent with the instructor's option to designate one dataset for the core session and reserve a second (e.g., Ames Housing) for the extended report.

2. Business Problem and Objective

2.1 Problem Statement

A real-estate analytics stakeholder requires a repeatable, data-driven method to estimate the market value of residential property from measurable characteristics of the property and its surrounding area. Framed formally, the objective is to learn a function $y = f(x_1, x_2, \dots, x_p) + \epsilon$ that maps a vector of property/district attributes to a continuous price target, and to evaluate that function on data withheld from the fitting process.

2.2 Central Question

How accurately, reliably, and transparently can property value be predicted from measurable property and location characteristics?

2.3 Stakeholders and Decision Context

The predictive model informs decisions across several stakeholder groups, each with a different sensitivity to prediction error:

- Property buyers and sellers - price discovery and negotiation support; large errors distort affordability decisions.
- Banks and mortgage lenders - collateral valuation checks; systematic overvaluation raises credit risk.
- Developers and investors - site selection and return estimation; biased predictions can misallocate capital.
- Public/tax agencies - mass appraisal and anomaly detection; transparency and consistency are essential.
- Real-estate platforms - automated valuation estimates; predictions must stay current and location-aware.

2.4 Target Definition and Units

The target variable in this study is the median house value of a California census block group, expressed in units of \$100,000 (e.g., a value of 4.526 corresponds to \$452,600). This is an important distinction from an individual sale price: the model predicts an aggregated, district-level statistic rather than the price of any single dwelling, and this distinction is carried through the interpretation and limitations sections of this report.

2.5 Success Criteria

- Technical - lower cross-validated and test RMSE/MAE than the naive mean-prediction baseline.
- Generalization - an acceptable, non-alarming gap between training and test performance.
- Diagnostic - no severe unexplained residual pattern and no data leakage.
- Business - errors expressed in the original target unit (\$100,000s) and discussed in context.
- Operational - a reproducible pipeline built from a fixed random seed and version-recorded libraries.

3. Dataset Description

3.1 Source and Provenance

Dataset: California Housing, loaded via scikit-learn's `fetch_california_housing(as_frame=True)` loader. This dataset derives from the 1990 U.S. Census and is distributed as a standard scikit-learn 'real-world' dataset for regression benchmarking.

3.2 Structural Summary

Property	Value
Rows (observations)	20,640 census block groups
Raw predictor columns	8 (all numeric)
Target column	Price (renamed from MedHouseVal)
Target unit	Median house value, \$100,000s
Missing values	None detected in any column
Duplicate rows	None detected
Train / Test split	16,512 rows (80%) / 4,128 rows (20%)
Random seed	42 (fixed for split and all stochastic models)

3.3 Data Dictionary

Feature	Description	Type
MedInc	Median income of the block group (10,000s USD)	Numeric
HouseAge	Median house age within the block group	Numeric
AveRooms	Average number of rooms per household	Numeric
AveBedrms	Average number of bedrooms per household	Numeric
Population	Total population in the block group	Numeric
AveOccup	Average number of household members	Numeric
Latitude	Block-group latitude	Numeric
Longitude	Block-group longitude	Numeric
Price (target)	Median house value, \$100,000 units	Numeric

3.4 Known Limitations of the Dataset

- Predicts a block-group median value, not an individual house sale price - a common conceptual error to avoid, as emphasised in the laboratory manual.
- Data reflect 1990 census-era prices and demographics; not current market conditions.
- Geographic scope limited to California; findings should not be generalized to other regions.
- Feature set is restricted to census-derived variables - no structural attributes such as garage, lot size, or renovation history are available.

4. Methodology

4.1 End-to-End Workflow

The study followed the ten-stage workflow specified in the laboratory manual: (1) business understanding, (2) data understanding, (3) holdout design, (4) data preparation, (5) exploratory analysis, (6) baseline modelling, (7) validation and tuning, (8) final evaluation, (9) interpretation and risk, and (10) reproducibility packaging.

4.2 Holdout Design

A random 80/20 train-test split (scikit-learn `train_test_split`, `random_state = 42`) was performed before any preprocessing was fitted, in order to prevent information leakage from the test partition into model training.

Because block groups are independent geographic units without repeated identifiers across rows, a random split was judged appropriate rather than a grouped or temporal split.

4.3 Preprocessing Pipeline

All eight numeric predictors were passed through a ColumnTransformer containing a StandardScaler, wrapped inside a scikit-learn Pipeline together with each estimator. Standardization is required for Ridge and Lasso because their penalty terms are scale-dependent, and a common pipeline structure was retained across all models for a fair, leakage-safe comparison. No categorical encoding was required because every feature in this dataset is numeric.

4.4 Cross-Validation and Hyperparameter Tuning

Ridge, Lasso, and Random Forest were tuned using GridSearchCV with 5-fold cross-validation (KFold, shuffle = True, random_state = 42), optimizing negative RMSE on the training partition only. The test partition was evaluated exactly once, after model selection was finalized, in accordance with the requirement that the test set must not be used repeatedly for model selection.

4.5 Reproducibility Controls

- Fixed global random seed (SEED = 42) applied to the train-test split and to all stochastic estimators.
- Library versions recorded at run time: NumPy 2.0.2, pandas 2.2.2, scikit-learn 1.6.1, Seaborn 0.13.2.
- Identical preprocessing pipeline structure applied to every candidate model to keep the comparison fair.
- Test-set evaluation performed once per model, after cross-validated tuning was complete.

5. Exploratory Data Analysis and Feature Engineering

Exploratory analysis was carried out on the training partition to avoid contaminating modelling decisions with information from the held-out test set.

5.1 Correlation Analysis (Figure 2 in notebook)

A correlation heatmap of all numeric features against the target confirmed that MedInc (median income) is by far the strongest linearly associated predictor of Price, with a Pearson correlation of 0.6906 on the training data. Latitude and Longitude show moderate, spatially structured relationships with price consistent with well-known coastal/inland value gradients in California, while Population and AveOccup show weak linear association with the target.

5.2 Feature-Target Relationships (Figure 3 in notebook)

Scatter plots with fitted trend lines were produced for MedInc, HouseAge, AveRooms, AveBedrms, Population, and AveOccup against Price. MedInc shows a clear positive, moderately linear relationship with visible

heteroscedasticity (spread increases with income). Several other features (AveRooms, AveBedrms, Population, AveOccup) show heavily compressed distributions with a small number of extreme outlier block groups, consistent with average-based aggregation over unevenly sized populations.

5.3 Target Distribution

The Price target is right-skewed, with a visible ceiling effect: the dataset caps median house values at 5.00 (\$500,000), causing a spike of censored observations at the upper bound. This censoring is a documented characteristic of the dataset and is discussed further in Section 9 (Limitations).

5.4 Observations Feeding Modelling Decisions

- Because MedInc has the strongest single-feature correlation with price, it was selected as the feature for the Simple Linear Regression baseline (Section 6.2).
- Because features vary substantially in scale (e.g., Population in the thousands vs. AveBedrms near 1), standardization was applied before any linear or regularized model.
- No missing-value imputation was required, since the dataset contains zero missing entries across all columns.
- The upper-bound censoring of Price was noted as a source of irreducible error for the highest-value block groups and is revisited in the residual and limitations discussion.

6. Model Development

Six models were implemented under the identical 80/20 split and evaluation protocol described in Section 4.

6.1 Model 1 - Naive Baseline

A `DummyRegressor(strategy='mean')` was fitted to establish the minimum bar that any learned model must clear. It always predicts the training-set mean price regardless of input features.

6.2 Model 2 - Simple Linear Regression

MedInc was selected as the single predictor because it showed the highest correlation with price (0.6906) among all eight features. The fitted equation was: $Price = 0.4446 + 0.4193 \times MedInc$

The positive slope (0.4193) indicates that, holding no other variable constant, each one-unit increase in median income (i.e., \$10,000) is associated with an expected increase of approximately \$41,900 in median block-group house value.

6.3 Model 3 - Multiple Linear Regression

All eight standardized features were used in an ordinary least-squares fit. This model captures the joint, conditional contribution of income, age, room counts, population density, and geography simultaneously, rather than relying on a single predictor.

6.4 Model 4 - Ridge Regression (L2)

Ridge regression was tuned over a logarithmic grid of 10 alpha values (10^{-3} to 10^3) using 5-fold cross-validation. The best cross-validated alpha was 2.1544, with a best CV RMSE of 0.7205.

6.5 Model 5 - Lasso Regression (L1)

Lasso was tuned over a logarithmic grid of 10 alpha values (10^{-4} to 10^0). The best cross-validated alpha was 0.0008, with a best CV RMSE of 0.7205. At this very small penalty strength, all eight coefficients remained non-zero, Lasso did not perform feature elimination on this dataset, indicating that every feature carries some independent predictive signal.

6.6 Model 6 - Random Forest Regressor

A Random Forest was tuned via grid search over $n_estimators \in \{100, 200\}$, $max_depth \in \{10, 15\}$, and $min_samples_leaf \in \{2, 4\}$. The best configuration was $n_estimators = 200$, $max_depth = 15$, $min_samples_leaf = 2$, achieving a best CV RMSE of 0.5175 - substantially better than any linear variant.

6.7 Model Summary Table

Model	Key Setting	Tuned?
Naive Baseline	strategy = mean	No
Simple Linear Regression	Feature: MedInc	No
Multiple Linear Regression	All 8 standardized features	No
Ridge Regression (L2)	alpha = 2.1544	Yes (GridSearchCV, 5-fold)
Lasso Regression (L1)	alpha = 0.0008	Yes (GridSearchCV, 5-fold)
Random Forest Regressor	n_estimators=200, max_depth=15, min_samples_leaf=2	Yes (GridSearchCV, 5-fold)

7. Evaluation Results

7.1 Final Test-Set Comparison

Model	MAE	RMSE	R ²	RMSE Improvement vs. Baseline
Naive Baseline	0.9061	1.1449	-0.0002	0.0%
Simple Linear Regression	0.6299	0.8421	0.4589	26.4%
Multiple Linear Regression	0.5332	0.7456	0.5758	34.9%
Ridge Regression (Tuned)	0.5332	0.7455	0.5758	34.9%
Lasso Regression (Tuned)	0.5332	0.7448	0.5766	35.0%
Random Forest (Tuned)	0.3304	0.5077	0.8033	55.7%

7.2 Cross-Validation Evidence

Model	Best CV RMSE (5-fold, train)
Ridge (tuned)	0.7205
Lasso (tuned)	0.7205
Random Forest (tuned)	0.5175

The close agreement between cross-validated RMSE and final test RMSE for every tuned model (e.g., Random Forest: 0.5175 CV vs. 0.5077 test) indicates that model selection was not overfit to the training folds and that the reported test performance is a trustworthy estimate of generalization.

7.3 Best Model

The tuned Random Forest Regressor is the best-performing model overall, with the lowest test RMSE (0.5077), lowest test MAE (0.3304), and highest test R^2 (0.8033), representing a 55.7% RMSE reduction relative to the naive mean baseline.

7.4 Diagnostic Evidence

Actual-vs-predicted and residual-vs-fitted plots were generated for the best model. Predictions track the identity line closely across the mid-range of prices, with visible compression at the upper end of the target range - consistent with the \$500,000 censoring ceiling discussed in Section 5.3. Residuals show mild heteroscedasticity, with slightly larger error spread for higher-value block groups, which is expected given both the censoring effect and the naturally larger absolute price variation among expensive properties.

8. Interpretation of Results

8.1 Feature Influence

Median income (MedInc) is confirmed, across both the simple linear model and the multiple/regularized models, as the dominant predictor of house value. Its correlation with price (0.69) is markedly stronger than any other feature, and it retains a large coefficient magnitude in the multiple regression and Ridge/Lasso fits. Geographic position (Latitude, Longitude) contributes secondary, spatially-structured signal, consistent with the well-documented value premium of coastal California relative to inland regions.

8.2 Why Regularization Made Little Difference Here

Ridge and Lasso achieved essentially the same test RMSE as ordinary Multiple Linear Regression (0.7455/0.7448 vs. 0.7456). This indicates that multicollinearity and coefficient instability were not severe problems for this particular feature set - the eight California Housing predictors are not strongly collinear, so there was limited variance for regularization to reduce. Lasso's very small optimal alpha (0.0008) and its retention of all eight non-zero coefficients corroborate this: no feature was redundant enough to be shrunk to zero.

8.3 Why Random Forest Outperformed the Linear Family

The Random Forest's substantially lower RMSE (0.5077 vs. ~0.745 for all linear variants) indicates that the true relationship between the predictors and price is meaningfully nonlinear and/or involves interaction effects (e.g., the effect of income on price may differ by location) that a linear model cannot represent. Ensemble tree methods can partition the feature space adaptively and capture such interactions without requiring the analyst to specify them in advance - at the cost of reduced coefficient-level interpretability.

8.4 Association, Not Causation

All reported relationships are conditional statistical associations estimated from observational, cross-sectional census data. None of the coefficients or feature-importance scores in this study should be interpreted as

evidence of a causal effect of, for example, income or room count on price; unmeasured confounders (e.g., local school quality, neighborhood desirability) plausibly drive both income sorting and price simultaneously.

8.5 Business-Facing Model Recommendation

Select Random Forest where predictive accuracy is the priority (e.g., automated valuation estimates); it provides the best generalization observed in this study, with a 55.7% RMSE improvement over the naive baseline. Select Ridge or Lasso where coefficient-level interpretability is required for stakeholder communication (e.g., explaining a valuation decision to a bank underwriter); both offer test performance within 2% of the full linear model while providing more stable, regularized coefficients. Before any deployment, the model should be accompanied by an explicit error range and an escalation rule for unusual or high-value properties.

9. Limitations and Responsible-Use Risk

9.1 Data Limitations

- The dataset predicts the median value of a census block group, not the sale price of any individual house - this distinction must be preserved in any downstream business use.
- Data reflect 1990 U.S. Census conditions; prices, incomes, and housing stock have shifted substantially since then, so absolute price predictions are not current-market-valid.
- The target variable is right-censored at \$500,000, compressing true variation among the most expensive properties and biasing predictions downward in that segment.
- Geographic scope is limited to California; the fitted relationships should not be extrapolated to other states or countries.
- Feature set is restricted to census-aggregated variables; no structural attributes (lot size, garage, renovation, school district) are available, which caps the achievable accuracy of any model on this data.

9.2 Modelling Limitations

- A single train-test split was used for final evaluation; a repeated-split or nested-CV protocol would give a tighter estimate of test-metric variability.
- Random Forest, while most accurate, offers weaker native interpretability and can extrapolate poorly outside the range of training data (e.g., unusually high-income block groups).
- Only one dataset (California Housing) was analysed in this submission; the laboratory manual's extended-study requirement calls for at least two datasets (e.g., adding Ames Housing) for a full comparative report noted here as a scope boundary of this core-lab deliverable rather than an oversight.

9.3 Responsible-Use Considerations

Housing price data can encode historical socioeconomic and spatial inequality. Any production use of location or income-linked features in a credit, taxation, or insurance context requires governance review, and predictions should never be treated as an unquestionable price authority. This model is a decision-support artifact from an academic exercise and is not validated for production valuation, lending, or taxation use without current data, uncertainty quantification, and human review.

10. Future Improvements

1. Extend the comparative study to a second, structurally different dataset (e.g., Ames Housing) containing mixed numerical and categorical attributes, satisfying the manual's two-dataset extended-study requirement.
2. Add Gradient Boosting / XGBoost as a further ensemble comparison, with reported training cost and interpretability trade-offs relative to Random Forest.
3. Apply a log-transform to the Price target to address right-skew and censoring, evaluating errors after inverse-transformation back to original units.
4. Introduce spatial cross-validation (e.g., grouping by geographic cluster) to test whether the current random split overstates generalization to genuinely new locations.
5. Add prediction-interval or quantile-regression estimates so that business users receive an error range alongside each point prediction.
6. Establish a monitoring plan to track model drift against updated housing-market data, since the source data is several decades old.
7. Investigate segment-level error (e.g., by income quartile or coastal vs. inland location) to check for unequal predictive performance across subgroups.

11. Conclusion

This study developed and evaluated six regression models for predicting median district house value (in \$100,000 units) using the California Housing dataset (20,640 block groups, 8 numeric predictors). The final selected model, a tuned Random Forest Regressor, achieved a test MAE of 0.3304, RMSE of 0.5077, and R^2 of 0.8033, improving RMSE by 55.7% relative to the naive mean baseline. The strongest predictive signal was median household income, with geographic position contributing secondary explanatory power; errors increased somewhat for the highest-value, right-censored block groups. The model is suitable for exploratory, decision-support estimation of district-level median value within California and within the historical data period represented, but should not be used for individual-property valuation, current-market pricing, or any credit/taxation decision without additional, more recent, and more granular data plus human review. The success criteria set out in Section 2.5 were met: the learned models comfortably beat the naive baseline (technical criterion), cross-validated and test RMSE remained closely aligned for every tuned model (generalization criterion), no unexplained severe residual pattern beyond the known censoring effect was found (diagnostic criterion), and all errors were reported in the original \$100,000 target unit (business criterion). The full workflow from holdout design through tuned final evaluation is reproducible via a fixed random seed and recorded library versions (operational criterion).

12. Discussion Questions

1. Why is a mean-prediction baseline necessary even when Linear Regression is available?

A naive baseline establishes the minimum acceptable bar: if a learned model cannot beat a model that simply predicts the training mean for every observation, the learned model adds no value. In this study the baseline RMSE was 1.1449; every subsequent model's improvement is reported relative to this floor (e.g., Random Forest's 55.7% RMSE reduction), which gives the improvement figures a concrete, defensible reference point rather than an arbitrary one.

2. What does a negative test R^2 imply?

R^2 compares a model's squared error to the squared error of always predicting the mean. A negative R^2 means the model performs worse than that trivial mean baseline on the evaluation data the naive baseline in this study recorded an R^2 of -0.0002 (essentially zero, as expected, since it is by definition the mean predictor evaluated on a slightly different partition than it was fit on).

3. Why may a feature have high correlation with price but a small multiple-regression coefficient?

A simple correlation ignores all other variables, whereas a multiple-regression coefficient measures the conditional association after holding the other included features constant. If a feature's apparent relationship with price is largely mediated through another correlated feature (for example, if Latitude/Longitude partly explain what looks like a Population effect), its multiple-regression coefficient can shrink even though its raw correlation is sizeable. This is precisely why coefficient interpretation must always be qualified by which other variables are in the model.

4. How can neighborhood information improve accuracy while also creating governance concerns?

Latitude and Longitude improved this study's models by capturing real spatial value gradients (e.g., coastal premiums). However, location is also a well-known proxy for historically embedded socioeconomic and racial patterns. Using location-linked features in models that influence credit, insurance, or taxation decisions risks perpetuating those historical inequities, so such use requires fairness auditing and governance review even when it measurably improves accuracy.

5. When would a temporal or grouped split be more realistic than a random split?

A temporal split is more realistic when the deployment use case involves predicting future prices from past data (e.g., forecasting next year's valuations), since a random split would let future information leak into training. A grouped split is preferable when the same property, neighborhood, or agent could appear in both partitions under random splitting, inflating apparent performance. In this study, a random split was judged acceptable because California Housing block groups are independent aggregated units without repeated identifiers, but a spatial/grouped split would be a natural robustness check (see Section 10).

6. Why must imputation, scaling, and feature selection be fitted only on training folds?

Fitting these steps on the full dataset (including test or validation data) leaks distributional information such as the test set's mean, variance, or missingness pattern into the training process, producing an overly optimistic performance estimate that will not hold on truly unseen data. In this study, StandardScaler was fitted only within the scikit-learn Pipeline on training folds during cross-validation and on the training partition for final fitting, keeping the test set genuinely unseen until final evaluation.

7. What evidence distinguishes heteroscedasticity from random residual variation?

Heteroscedasticity shows a systematic pattern residual spread that visibly widens or narrows as a function of the fitted value or a specific predictor whereas random variation shows a roughly constant residual band across the full range of fitted values. In this study's residual-vs-fitted plot, spread widened noticeably for high predicted prices, a systematic pattern (driven partly by target censoring at \$500,000), which is evidence of heteroscedasticity rather than pure random noise.

8. Why can Random Forest improve accuracy but weaken extrapolation and coefficient-style interpretation?

Random Forest predicts by averaging many decision trees, each of which partitions the feature space into regions and predicts a constant value within each region. This lets it flexibly capture nonlinearities and interactions (explaining its RMSE advantage here), but it cannot extrapolate beyond the range of values seen during training a tree-based prediction for a block group with an income far outside the training range will

simply be pinned to the nearest training region's value. It also has no single global coefficient for a feature; 'importance' is a much coarser, less directly interpretable summary than a regression slope.

9. How does target log transformation change coefficient interpretation and error behavior?

Under a log-target transform, a linear coefficient is interpreted approximately as a percentage change in price per unit change in the predictor (rather than an absolute dollar change), and errors become more homogeneous in relative terms because large, skewed absolute values are compressed. This was identified as a promising future extension (Section 10) for this right-skewed, censored target.

10. What practical evidence is needed before using the model for mortgage or taxation decisions?

At minimum: current (not 1990-era) training data; documented, quantified error ranges rather than point estimates alone; segment-level error analysis (to confirm errors are not systematically worse for particular income, age, or location groups); a fairness/bias audit of any location-linked features; and a monitoring plan with defined retraining triggers. None of these are currently in place for this academic model, which is explicitly stated in the Limitations section.

11. If two models have nearly identical RMSE, what other criteria should determine selection?

In this study, Multiple Linear Regression, Ridge, and Lasso all produced RMSE within 0.001 of one another (0.7456, 0.7455, 0.7448). With performance effectively tied, the deciding factors become interpretability (Ridge/Lasso give more stable, business-explainable coefficients), computational cost (linear models train far faster than Random Forest), robustness to future data drift, and ease of deployment/monitoring — not marginal RMSE differences that may simply reflect split-to-split noise.

12. How would inflation and market drift affect a model trained on historical housing data?

Because the California Housing data reflect 1990 census conditions, absolute price predictions from any model trained on it — including the tuned Random Forest in this study — will be systematically stale relative to today's housing market. Both the price level (inflation) and the relative importance of features (market drift, e.g., changing preferences for room count or location) can shift over decades, meaning the model's coefficients and error behavior would need to be re-estimated on current data before any real-world price forecasting use.

13. Viva Questions

1. Define supervised learning and regression.

Supervised learning uses a dataset of labelled input-output pairs to learn a function that maps inputs to outputs. Regression is the branch of supervised learning used when the output (target) is continuous — in this study, the continuous median house value.

2. Distinguish simple from multiple Linear Regression.

Simple Linear Regression uses exactly one predictor (MedInc in this study); Multiple Linear Regression uses two or more predictors simultaneously (all 8 features here), allowing it to model the joint, conditional effect of several variables at once.

3. What is a residual?

A residual is the difference between an observed value and the model's predicted value for that observation: $\text{residual} = \text{actual} - \text{predicted}$. Residuals were computed and plotted for the best model in Section 7.4.

4. Why does OLS square the residuals?

Squaring makes all residuals non-negative (so positive and negative errors don't cancel out), penalizes larger errors more heavily than smaller ones, and yields a differentiable, convex objective function with a closed-form analytical solution.

5. State the interpretation of a regression coefficient.

A coefficient represents the expected change in the predicted target for a one-unit increase in that predictor, holding all other included predictors constant. For the Simple Linear Regression model in this study, the MedInc coefficient of 0.4193 means each \$10,000 increase in median income is associated with roughly a \$41,900 increase in predicted median house value.

6. What is data leakage? Give one example from this lab.

Data leakage occurs when information from outside the training data (typically from the test set or the future) improperly influences model fitting, inflating apparent performance. In this lab, leakage was avoided by performing the train-test split before fitting the StandardScaler; had the scaler instead been fitted on the full dataset before splitting, test-set statistics would have leaked into the training process.

7. Why is the test set kept untouched during tuning?

Repeatedly evaluating and selecting models based on test-set performance effectively turns the test set into another validation set, biasing the final reported performance upward. All tuning in this study (Ridge, Lasso, Random Forest) used only 5-fold cross-validation on the training partition; the test set was scored exactly once, after the best hyperparameters were already chosen.

8. Compare MAE and RMSE.

MAE (Mean Absolute Error) averages the absolute residuals and is less sensitive to a few large errors. RMSE (Root Mean Squared Error) squares residuals before averaging and then takes the square root, which penalizes large errors more heavily. In this study, RMSE was consistently higher than MAE for every model (e.g., Random Forest: MAE 0.3304 vs. RMSE 0.5077), indicating the presence of some larger-than-typical errors, likely concentrated among the high-value, censored block groups.

9. Can R^2 be negative? Explain.

Yes, R^2 can be negative on unseen data if the model's squared error exceeds the squared error of simply predicting the mean of that data. This occurred (essentially, $R^2 \approx 0$) for the naive baseline in this study, which by construction cannot outperform its own definition.

10. Why is scaling important for Ridge and Lasso?

Ridge and Lasso penalize the magnitude of coefficients directly. If features are on very different numeric scales (e.g., Population in the thousands vs. AveBedrms near 1), the penalty would unfairly shrink coefficients of large-scale features more than small-scale ones purely due to units, not true importance. StandardScaler was therefore applied to all 8 features before both penalized models.

11. How do Ridge and Lasso differ?

Ridge (L2) penalizes the sum of squared coefficients, shrinking all coefficients smoothly toward zero without eliminating any. Lasso (L1) penalizes the sum of absolute coefficients, which can shrink some coefficients exactly to zero, performing implicit feature selection. In this study, Lasso's optimal alpha (0.0008) was small enough that all 8 coefficients remained non-zero - no feature was eliminated.

12. Why might Lasso set some coefficients to zero?

The L1 penalty's geometry (a diamond-shaped constraint region) has corners aligned with the coordinate axes, so the optimal solution often lands exactly on a corner where one or more coefficients are exactly zero effectively removing that feature. This did not occur strongly in this dataset because the optimal alpha was very small, suggesting all 8 features carry independent signal.

13. Is Polynomial Regression a linear model? Explain.

Yes, Polynomial Regression is linear in its coefficients (it is still fit by ordinary least squares on the expanded feature set), even though it is nonlinear in the original predictor. For example, $y = \beta_0 + \beta_1x + \beta_2x^2$ is linear in $\beta_0, \beta_1, \beta_2$. This study used only linear and tree-based models; polynomial regression is listed as a possible future extension.

14. What is multicollinearity and how can it affect coefficients?

Multicollinearity occurs when predictors are highly correlated with one another, making it difficult for OLS to attribute the target's variation uniquely to any one of them; coefficients can become unstable, inflate in magnitude, or even flip sign between similar samples. In this study, Ridge and Lasso performed almost identically to plain Multiple Linear Regression, suggesting multicollinearity among the 8 California Housing features was not severe.

15. What is heteroscedasticity?

Heteroscedasticity is non-constant residual variance across the range of fitted values - for example, errors that grow larger for higher predicted prices. The residual-vs-fitted plot for the best model in this study showed exactly this pattern at the high end of the price range, linked to the dataset's \$500,000 value ceiling.

16. How do you detect overfitting from train and test metrics?

Overfitting is suspected when training error is much lower than test/validation error (the model has memorized training-specific noise rather than learning generalizable structure). This study compared train-vs-test metrics for Linear Regression, Ridge, and Random Forest; the close alignment between cross-validated and test RMSE for the tuned Random Forest indicated healthy generalization rather than overfitting.

17. What does cross-validation estimate?

K-fold cross-validation estimates how a model is expected to perform on unseen data by repeatedly training on K-1 folds and validating on the held-out fold, then averaging the resulting metric. In this study, 5-fold cross-validation was used during GridSearchCV to select hyperparameters without touching the final test set.

18. Why can a Decision Tree fit nonlinear relationships?

A Decision Tree partitions the feature space using a sequence of axis-aligned splits (e.g., 'MedInc > 4.2?'), producing a piecewise-constant prediction surface that can approximate arbitrarily complex, nonlinear relationships given enough splits, without requiring the analyst to specify a functional form in advance.

19. How does Random Forest reduce variance?

Random Forest trains many decision trees on bootstrap-resampled subsets of the data (and random feature subsets at each split), then averages their predictions. Averaging many diverse, individually high-variance trees reduces the overall variance of the ensemble prediction, which explains its markedly lower RMSE (0.5077) compared to a single linear model in this study.

20. Why should model errors be analyzed by price segment?

Aggregate metrics like RMSE can hide unequal performance across subgroups a model might perform excellently for mid-priced homes but poorly for very cheap or very expensive ones. This study's diagnostics

flagged wider residual spread at the high-value end (linked to censoring), a segment-level pattern that a single overall RMSE figure alone would not reveal.

21. What is the role of a pipeline in scikit-learn?

A Pipeline chains preprocessing steps (e.g., StandardScaler) and an estimator into a single object that can be fit and cross-validated as one unit. This guarantees that preprocessing is refit correctly on every training fold and never leaks information from validation/test data the core mechanism used throughout this study to keep all six models leakage-safe and comparably preprocessed.

22. What is the difference between prediction and causal inference?

Prediction aims only to estimate the outcome accurately from observed inputs, regardless of why the relationship holds. Causal inference aims to estimate the effect of deliberately changing one variable on the outcome, which generally requires additional assumptions (e.g., randomization, controlling for confounders) beyond what an observational regression like this study provides. All findings here are associational, not causal.

23. Why can a model fail when moved to a different city or year?

A model trained on California, 1990-era data has implicitly learned relationships (price levels, feature importance, income distributions) specific to that place and time. Moving it to a different city or year invokes distribution shift the new data's feature-target relationship may differ substantially, degrading accuracy unless the model is retrained or recalibrated on local, current data.

24. What information should accompany a saved model artifact?

A saved artifact should be accompanied by: the fixed random seed and library versions used, the exact feature set and preprocessing steps (ideally packaged inside the same pipeline object), the training/test split sizes, the final hyperparameters, the recorded evaluation metrics, and a note on the model's valid scope and known limitations all of which are documented for the Random Forest pipeline in this report.